

WASP-2b

R-0159

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-2_180916 · created 2026-07-16T15:17:20+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2458377.7527 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.155 +/- 0.011
Transit depth (Rp/Rs)^2	2.39 +/- 0.35 [%]
Orbital Inclination (inc)	85.09 +/- 0.65
Airmass coefficient 1 (a1)	0.986 +/- 0.014
Airmass coefficient 2 (a2)	0.0123 +/- 0.0088
Scatter in the residuals of the lightcurve fit is	1.38 %
Best Comparison Star	#3 - [483, 156]
Optimal Aperture	4.7
Optimal Annulus	8.11
Transit Duration (day)	0.0788 +/- 0.0062

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.155 $\pm$ 0.011 vs 0.1326 (deviation 1.94 σ)
Duration fit vs published	0.0788 d vs 0.0737 d (deviation 0.77 σ)
Mid-time O-C	12.9 min
Depth SNR	13.4
Residual scatter	1.38 %

Light curve

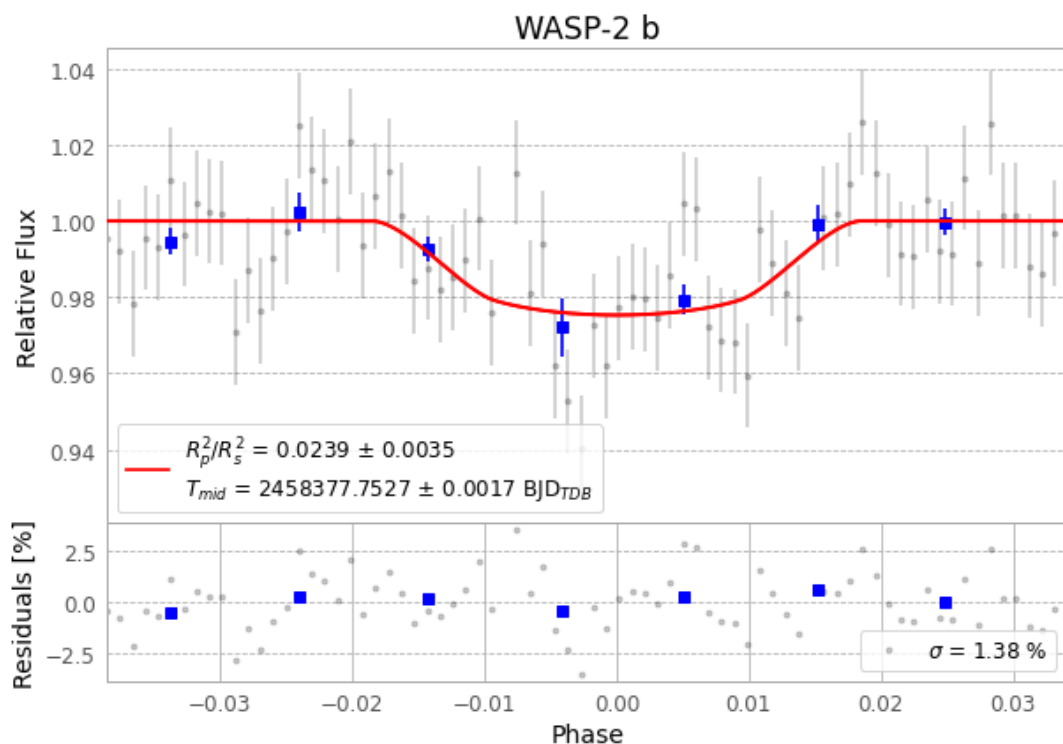


Figure 1 — FinalLightCurve_WASP-2 b_2018-09-15.png (SHA-256 7c3a1284c48e5ec6...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [289, 219], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5d3c0ad3b500eebe...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit c71eff8

Data provenance

76 FITS frames (50 MB), WASP-2180916035712.FITS → WASP-2180916074212.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-2-180916.log (aecf0222ca37...), FinalLightCurve_WASP-2 b_2018-09-15.png (7c3a1284c48e...), FinalLightCurve_WASP-2 b_2018-09-15.csv (d519956fa4fa...)

Compute resources

Wall time 120.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-16 15:17Z.